In this stage this file is very short. In the future I will turn it into a proper user manual explaining the software and how to use it in great detail. As of right now the software is just not ready for that. Right now you are reading an instruction that will just get you to mill your first pcbs.

If you properly finished the assembly guide when launching the python program you should be in the main menu and the program should be connected to the machine and two cameras.

As of right now even I don’t use the full capabilities of the python software I wrote for this machine. I use it to generate gcode but I send it to the machine using Universal Gcode Sender. The PnP function of the python program sorta works most of the time but it still in a very early stage. I have a few ideas how to greatly improve it in the future and I’m actively working on it.

As of right now the workflow for making 1 layer PCBs is as follows:
- Generate gerber files in your cad software
- If they are in a zip format extract it
- Launch the python program and get to the main menu
- Press the option 1
- Choose the proper gerber files you want to mill (top layer, holes, vias, cutout etc.)
- Save the generated files
- Either quit the program and send those files to the machine using UGS or continue using the python script

No matter what option for sending the gcode files you choose you still need to mount the pcb to the bed somehow. To do so take the pcb holders, and put one on at least each corner of the pcb, then tighten the screw so that it holds it down. Then add one additional holder and in between it and the pcb put the pcb probe so that when the mill with the clamp touches it during autoleveling it will close the circuit. Also right before milling you need to put some sort of coolant on the pcb. I use WD40. Reapply it while the machine is milling if you see it has dried up. Also before your first run, keep a hand near the power switch and watch the first pass closely - a wrong Z-zero or bad autolevel can crash the bit into the bed.

If you choose to use UGS to send the gcode files you need to manually move the machine to the point 0 and reset the xyz coordinates to 0, probe the z level and do the autoleveling.

If you choose to use the python program instead make sure you calibrated everything correctly (option 3 from the main menu). If you did the program should automatically detect where the pcb is on the bed, move there, probe the Z height and do the autoleveling. Then it will mill the pcb. If you follow the instructions on the screen you should end up with a properly milled pcb.